Machine Learning Project - Heinz College 90803 S25
Project: Predicting worst performing projects listed on DonorsChoose
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Executive Summary

Public school funding inequities disproportionately affect schools in high-poverty districts. These schools often receive fewer resources per student, leading to lower student performance, reduced student retention, and limited economic mobility in the long term. Platforms like DonorsChoose aim to mitigate these disparities by enabling teachers to request donations for classroom projects. However, a critical limitation remains that only fully funded projects receive donations. Campaigns that fall short of their goals get no funds, further exacerbating funding gaps. Our project addresses this issue by identifying campaigns least likely to reach their funding goals early in the donation cycle, allowing for targeted intervention.

Specifically, we built a machine learning pipeline to predict the bottom 10% of projects most at risk of not being fully funded. We then send the predictions to DonorsChoose digital content experts for review. We trained and evaluated several models-including Stochastic Gradient Descent (SGD) with Logistic Regression, Random Forest, and XGBoost- on project data from 2003 to 2012. The XGBoost model achieved the best performance, with an F1-score of 90%, making it the most effective for early detection.

The model shows that school poverty level, donor engagement, and teacher affiliation were consistently highly correlated with lower likelihood of project success. Over 80% of the least likely to be funded projects came from schools where more than 65% of students qualified for free or reduced-price lunch. Additionally, projects with below-average donor interaction, measured by the percentage of unique donor comments , were significantly less likely to succeed. Furthermore, projects led by teachers not affiliated with *Teach for America* were more frequently underfunded. This finding suggests that institutional network effects may influence donor behavior. Secondary factors such as project location and the material requested also accounted for some of the variability in project funding. Based on this analysis, we recommend:

1. **Increasing Visibility of At-Risk Projects on Social Media to Retain More Donors:** DonorsChoose should spotlight projects predicted to not reach their goals, particularly from high poverty schools. They should also encourage teachers to amplify the social media posts to increase donor engagement.

2. **Consolidating Project Submissions at the School Level to Maximize Donations:** Behavioral Economicss studies find that too much choice can lead to confusion and increase choice deferral. DonorsChoose should implement a system that allows donors to select a school rather than individual teachers projects. This strategy can reduce overload, mitigate biases related to teacher affiliation, and increase donations.

3. **Reforming Public School Funding formulas at the State and Federal Levels to Reduce Inequities:** Individual donations can reduce funding gaps between schools in the short term. However, these systemic issues require that state and federal governments

change the funding formulas to ensure resources are allocated based on community and students needs.

By leveraging predictive modeling and policy changes, DonorsChoose and education stakeholders can make meaningful progress in reducing public school funding disparities.

Background and Introduction

The Problem being solved

Public schools funding in the U.S. is inequitable across and within states.[1] Furthermore, high-poverty districts get less funding per student than low-poverty districts.[2] The disparity leads many teachers to fund classroom projects out of pocket. DonorsChoose is an online platform that allows teachers to seek funding and not rely solely on their finances. Although over 2 million projects have been funded through DonorsChoose, 1/3 of the projects still go unfunded. Our project aims to maximize the number of projects funded to ensure as many students as possible can access quality education.

Why is this problem important?

Teachers' access to more funds to buy school supplies and lead impactful projects like school trips can have far reaching effects, including the retention of young students in school, especially students from historically marginalized communities. Many studies find that districts with more resources focused on student learning tend to have higher education equality and better student outcomes such as performance in test scores.[3] This effect mainly comes from the fact that richer districts can allocate significant resources to schools, allowing them to have smaller class sizes, better facilities, and access to richer cultural experiences.[4] Retaining young students in school, especially those from more impoverished backgrounds can have a long lasting impact on their lives as well as their families.

What is the solution's impact?

The public school system can increase social mobility through equalizing opportunities for children from various backgrounds.[5] Increasing teachers' access to funding can act as an equalizer, reducing the wide gap in educational attainment between kids from rich and impoverished families. Increased funding can also increase students' earnings in the future. A study from the Brookings Institution finds that an additional year in school is associated with a

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[1] Allegretto, Sylvia, Emma García, and Elaine Weiss. "Public Education Funding in the U.S. Needs an Overhaul: How a Larger Federal Role Would Boost Equity and Shield Children from Disinvestment during Downturns." Economic Policy Institute. Accessed April 12, 2025. https://www.epi.org/publication/public-education-funding-in-the-us-needs-an-overhaul/.

[2] Ibid.

[3] Hartman, William T. "Education Funding Disparities: What Do the Dollars Buy?" *Journal of Education Finance* 24, no. 3 (1999): 389–408. http://www.jstor.org/stable/40704073.

[4] Lampl, Peter. "Unequal School Funding Hobbles the American Dream." *Financial Times*, October 17, 2024, sec. Education. https://www.ft.com/content/28e923e2-d934-4aa0-8cbe-2d6f4b2f4014.

[5] Reeves, Richard V. "Funding Gaps in Public Schools Real Problem for Social Mobility, Not Parents' Giving." Brookings. Accessed April 12, 2025. https://www.brookings.edu/articles/funding-gaps-in-public-schools-real-problem-for-social-mobility-not-parents-giving/.

10 percent increase in wage.[6] Therefore, in the short term, our solution can retain more kids in school, and in the long term improve their livelihood through increased wages.

Who will take action?

DonorsChoose's digital content expert will review the projects least likely to be funded before they expire and subsequently receive no funds at all. Besides review, DonorsChoose could also conduct research about the impact of funded projects on student outcomes compared to the projects that do not get funded. Findings from this research can convince more people to donate as they are able to see the value of their donations. DonorsChoose can also share the findings with education organizations that advocate for increased funding for poorer districts so they can highlight the importance of making school funding more equitable across the country.

Policy Goals

Effectiveness Goals: The project aims to maximize the donations on the DonorsChoose platform so that no dollar is lost. Projects that do not reach their target receive no funding. Instead of losing that money, our model will allow us to prioritize those projects and highlight them on the platform to increase their probability of success..

Efficiency Goals: We aim to reduce operational costs for DonorChoose by lowering the time needed to review which projects are likely to not get funding. Consequently, the digital content expert can make decisions more quickly about how to allocate resources.

Equity Goals: We highlighted earlier that funding for schools vary widely across states and districts depending on their poverty levels. Donors are more likely to contribute to projects from districts they identify with. Poorer people are less likely to donate given their financial constraints. Therefore, we also aim to provide projects from lower-income districts with more opportunities to be funded by highlighting them earlier in the review process.

Current and Previous Solutions

The disparity in public education funding has been and is being addressed through multiple pathways. For instance, Pledge 1% is a nonprofit that has encouraged 18,000 companies worldwide to commit a percentage of their assets to creating social impact.[7] Through this movement, many organizations such as Salesforce have committed part of their revenues to funding education programs especially in disadvantaged areas.[8] Parent-Teacher associations and local fundraising also represents another solution to the disparity in funding, though unsustainable.[9] Across the country, parents, teachers, and grassroot organizations have also been advocating for more equitable funding in U.S. public schools.[10] Between 1971 and 2010, state

[6] Kearney, Melissa S., and Phillip Levine. "Inequality Undermines the Value of Education for the Poor." Brookings. Accessed April 12, 2025. https://www.brookings.edu/articles/inequality-undermines-the-value-of-education-for-the-poor/.

[7] Pledge 1%. "Our Impact." *Pledge 1%* (blog). Accessed April 12, 2025. https://www.pledge1percent.org/our-impact/.

[8] Salesforce. "Commitment to Education - Grantee Spotlights." Salesforce. Accessed April 12, 2025. https://www.salesforce.com/company/commitment-to-education/grantee-spotlights/.

[9] Reeves, Richard V. "Funding Gaps in Public Schools Real Problem for Social Mobility, Not Parents' Giving." Brookings.https://www.brookings.edu/articles/funding-gaps-in-public-schools-real-problem-for-social-mobility-not-parents-giving/.

[10] Juliana. "Communities Are Leading the Call for Education Funding Reform So That All Students Can Thrive." NEA Foundation, March 27, 2024. https://www.neafoundation.org/ideas-voices/communities-are-leading-the-call-for-education-funding-reform-so-that-all-students-can-thrive/.

supreme courts mandated school finance reforms in 28 states to reduce inequities in public school funding.[11]

ML Solutions before ours

Dirisam et al. (2021)[12]: This study uses machine learning to predict the 10 most important factors that lead a donor to pick a project on the DonorsChoose platform. Dirisam et al. motivate this research by highlighting that low donor retention is a major challenge to projects being funded on the DonorsChoose platform. Only 20-25% of donors are interested in making more than one donation. The authors attempt to predict donor behavior so teachers can better formulate their project proposals and motivate one-time donors to give again. Dirisam et al. use a number of machine learning models to make their predictions, including Random Forest, Logistic Regression, and Naive Bayes. The authors find that Random Forest provides more than 60% of accuracy but does not work well with a larger number of features and data points. The study also shows that logistic regression and Naive Bayes improve precision scores above 70%. Dirisam et al. conclude that projects with high level of description, and features like project-resource-summary, and datetime show promising results in attracting donation.

Alazazi et al. (2020)[13]: This study explores the driving factors of success of campaigns posted on crowdfunding platforms. The authors motivate the research through emphasizing that funding significantly varies across projects on donation platforms. For instance, only 44% of projects posted on Kickstarter get fully funded. Kickstarter highlights creative projects, including education related campaigns. The authors predict how features such as social media shares can impact the amount of daily funding using supervised machine learning models such as SVM, K-nearest neighbors, and Random Forest. The authors find that the SVM algorithm performed well to predict amount of funding using number of days since the project was posted. Additionally, The performance of all their 6 models improved after the addition of features such as social media shares, likes, and updates. The authors recommend that crowdfunding platforms promote on their websites projects that were posted a long time ago but did not receive enough funding. Exposure can increase donation once enthusiasm wears off.

Zhang et al. (2019)[14]: This research explores project funding for 5000 small and medium innovative tech companies. The authors motivate their study by showing that in the U.S., less than ¼ of projects submitted to the National Science Foundation. In China, less than 22% of applicants get their projects funded by the National Natural Science Foundation. Using machine

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11 Jackson, C. Kirabo, Rucker C. Johnson, and Claudia Persico. "The Effects of School Spending on Educational and Economic Outcomes: Evidence from School Finance Reforms." Working Paper. Working Paper Series. National Bureau of Economic Research, January 2015. https://doi.org/10.3386/w20847.

12 Dirisam, Joel Varma, Doina Bein, and Abhishek Verma. "Predictive Analytics of Donors in Crowdfunding Platforms: A Case Study on Donorschoose.Org." In 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC), 0834–38, 2021. https://doi.org/10.1109/CCWC51732.2021.9376094.

13 Alazazi, Massara, Bin Wang, and Tareq Allan. "Success Factors of Donation-Based Crowdfunding Campaigns: A Machine Learning Approach." *Hawaii International Conference on System Sciences 2020 (HICSS-53)*, January 7, 2020. https://aisel.aisnet.org/hicss-53/dsm/data_mining/9.

14 Zhang, Chuqing, Han Zhang, and Xiaoting Hu. "A Contrastive Study of Machine Learning on Funding Evaluation Prediction." *IEEE Access* 7 (2019): 106307–15. https://doi.org/10.1109/ACCESS.2019.2927517.

learning, they seek to predict projects unlikely to get funded so applicants can better prepare their proposals. The authors used five supervised machine learning models, including decision tree, logistic regression, and SVM to predict funding decisions. They find that SVM models are best for funding prediction in terms of accuracy and F-1 score. However, the authors highlight that models such as logistic regression are more sensitive to unbalanced data and large sample sizes.

Theoretical Contribution

The literature review indicates that machine learning models such as SVM, Logistic Regression, and Random Forest can predict with high accuracy the factors that are most likely to attract donation, the amount of daily funding a project will get, and the probability that a project gets picked for research funding. The studies find that features like project description, posting date, and social media engagement can help increase model performance. Our study adds to this literature by applying machine learning models to predictions not yet explored. We identify the top 10% of projects that are least likely to get funding and submit them for review. Compared to other work in this field, our predictions are more targeted and can help DonorsChoose experts maximize their resources by focusing on a specific set of projects to review. Additionally, most of the literature focuses on smaller datasets. Our project uses larger datasets and make predictions on unbalanced data. Overall, our work complements the studies reviewed. After we identify the projects least likely to receive funding, findings from previous research can be used to help these projects reach their goals. For instance, DonorsChoose can apply learnings from Alazazi et al. such as increasing website exposure of the projects identified to attract donation.

Problem Formulation and Overview of Solution

DonorsChoose gives access to teachers to platforms that allow them to post funding requests for 'projects'. As we see from the data, these projects are usually asking for funding for stationery supplies, educational visits etc. The basic problem is that only some projects are funded - as funding is finalized only for those projects who cross a certain threshold. Often, projects are not funded because they don't cross the threshold. The problem at hand is to alleviate the project listings that **do not** receive the threshold funding. The given arrangement is the use of an expert in content formulation - who will work on these under-funded projects to help them cross the threshold, and thereby receive full funding.

The machine learning problem thus is simply to be able to correctly predict the problems that will not receive funding. Then, considering the limited scope of resource (content analyst expertise) - we will select and finalise the least probable projects in terms of full funding. Once we identify the least probable decile in terms of full funding possibility, DonorsChoose will work with an expert to alleviate the project - so that it meets the threshold for full funding. That is the larger goal.

This project, if able to predict reliably, can be a lighthouse project. Education policy, as well non-profits in the funding space can replicate this to their generalized problems of predicting if new instances in their own settings are poised to receive their required funding. This will enable the redirecting of scant resources to the projects that need them - so that resources are efficiently utilised for maximum output of preference (here, being full funding of as many projects as possible).

Data Description

The dataset we have used is relatively old. It is between 2003-13. We are predicting the status of projects in 2013 as part of the test. So while old, the dataset is big enough and helpful for the year we are working on.
We had other findings based on the data -

The typical total donations were capped at 1500(units) with an average of 75 (units) students being impacted by the donation. The total donation amounts over the years has been on the rise from 2003 to 2013 with occurring in 2008, 2011 and 2013. 2011 experienced the biggest drop in terms of the total donation. The number of projects funded generally increased from 2003 to 2013, and likewise unfunded projects. However, the number of unfunded projects in 2012 dropped compared to before the year. Projects from regions labelled as “high poverty” had better fundings relative to projects labelled as “low poverty”. Supplies, technology and books were the most funded resource types and comparatively trips, visitors and others being the least funded. Projects from New York and California were the most likely to be funded. Projects from Florida, Illinois, Indiana, North Carolina, South Carolina and Texas ranked second in terms of funding likelihood relative to other states. This was also the function of the fact that these states also had greater numbers of project postings in general. There are missing values for 17 out of the 34 features available. This includes but is not limited to the donor_city, donor_state, donation_message.

Our Work

GitHub link: [***https://github.com/davforson/DonorsChoose-Project***](https://github.com/davforson/DonorsChoose-Project)

Pre-processing

Five different datasets - namely, donations.csv, essays.csv, outcomes.csv, projects.csv and resources.csv - were merged using their unique project identifiers to create a common file for the analysis on the DonorsChoose projects. This file was then cleaned, and columns with null values were reviewed. We dropped 17 columns via discussions and collective decision making. These were dropped for the main reasons - 1) for lack of relevance to what we were trying to predict (for example, teacher_acctid and item_number), and 2) due to correlated details - ‘short_descriptions’, ‘need_statement’ etc. These columns were also descriptive data types, and considering the timelines, we decided to drop them and rely on objective datatype columns - categorically 1 and 0 - ‘great_chat’. The understanding we have is that it would be representative of the nature of the project stakeholders and works as a proxy to the qualitative datatype columns we dropped.

We split the data during exploratory data analysis - to understand the balance of our prediction variable. Amongst other insightful findings about the data set, EDA helped identify that our data was imbalanced. For this, we initiated SMOTE (Synthetic minority oversampling technique) over our dataset. The purpose of this was to uplift the minority class, in our case fully_funded = 0, to give balance to the dataset. The algorithm synthetically generated minority samples using internal computations based on the dataset. This resulted in a balanced dataset.

Next, we constrained our dataset to 15,000 instances by systematic sampling. We tried running our training on the entire dataset, but it didn’t work and we faced technical stoppages. It ran on 10,000 datapoints and 15,000 datapoints. We have used systematic sampling, which divided the dataset into windows such that the total windows were 15,000. The package then used one observation from each window - or every 15th observation to make up the sample data set. Following this, we imputed null values with the median - considering how our data set was now 'balanced' and the median would be representative of the central tendency. Following this, we deployed the OneHotEncoder, to obtain binary values of our variables. This concluded our pre-processing and we trained our models on this dataset.

Seeing how this is a time-series dataset, and our object was to classify a new instance, we split our dataset into training and testing sets via a sliding window mechanism. For the first instance of splitting, we divided the dataset into 2003-2010 - as a training set. We deployed this over 2011 datapoints as a testing set. This helped us obtain the baseline performance - apart from the simple idea that we knew about - and how around 21% of all projects are not funded. We are measuring the f-1 score as our metric of choice. The baseline performance we measured in 2011 was 0.90. For the second instance of splitting, we divided the dataset into 2003-2012 - as a training set. We deployed this over 2013 datapoints as a testing set. The f-1 score for this was also 0.90.

Reasoning for the train-test split to be divided in this manner was due to the fact that over 2012-13, lots of new data instances were created - either as a result of better data collection, or due to a more innate behavioural pattern in terms of greater interest in funding projects for school and more interest in education access. For example, 2012-13 have additional states - that didn't have projects before being represented. There are more unique categories in the 2012-13 part of the data set than there are in the 2003-2011 dataset for the different columns. To be able to sample in such unique categories, and to be able to instil in the model training that allows it to capture such changes in the testing sets, and more generally in the real world, we split the sliding windows this way.

Model Mechanics

Machine Learning methods: Three different machine learning models were used for analysis: Stochastic Gradient Descent(SGD) Classifier with Logistic Regression, Random Forest Classifier and XGBoost Classifier.

The SGD Classifier is a linear model optimized through stochastic gradient descent. Its main strengths lie in its scalability to large and high-dimensional datasets, fast training time, and ease of interpretation due to linear coefficients. It can be unsuitable for capturing complex and non-linear patterns, and is sensitive to hyperparameter tuning. It performs less ideally on imbalanced datasets, but our SMOTE run controlled for that so we tried this just the same. We found our model returns different accuracies on different runs.

The Random Forest Classifier, is an ensemble of decision trees trained using bootstrap aggregation (bagging). It works up a strong balance between performance and robustness. It effectively captures non-linear relationships, is relatively easy to tune, and is less prone to overfitting due to averaging across trees. Random Forests also offer insight into feature importance and are generally stable across runs. However, they are typically less accurate than boosting methods like XGBoost, and they can be memory-intensive with slower inference times in large-scale deployments.

In contrast, the XGBoost Classifier is a gradient boosting algorithm that builds an ensemble of decision trees in a sequential manner to minimize errors. It is good at modeling non-linear feature interactions and normally achieves high predictive accuracy, especially on structured data - which ours largely was. XGBoost includes built-in regularization, handles missing values - which we managed in pre-processing itself. Despite these advantages, it requires significant computational resources, and so we could limit our training only to 15,000 instances. Overfitting can still be a concern if not managed with cross-validation, in our case using sliding windows. The best performing method was found to be XGBoost, and we saved this trained model as ‘model_pickle’. XGBoost performed the best with an f-1 score of 0.90 for both 2011 predictions and 2013 predictions. All the following work was done for 2011 and 2013 using ‘model_pickle’ - our saved model.

Hyperparameters:
For SGD Classifier, the alpha, maximum iteration, loss, penalty, tolerance and random_state parameters were optimized using the GridSearchCV for reproducibility and compute optimization apart from the method’s requirement like regularization.
For both the Random Forest and XGBoost Classifiers, the number of estimator parameters were optimized.

Features used:
The features used were: is_exciting, donor_state, school_state, at_least_1_teacher_referred_donor, vendorid, great_chat, school_charter, teacher_teach_for_america, primary_focus_subject, resource_type, poverty_level, grade_level, month, ‘year’, ‘fully_funded’. Two of these features(month, year) were engineered from the ‘date_posted’ feature.

Tools:
sklearn, numpy, pandas, seaborn, imblearn(SMOTE), matplotlib, pickle

Bias auditing:
Conducting a comprehensive bias audit in the context of predicting whether educational projects will be fully_funded = 0 is essential to evaluate how the model performs across different sensitive or potentially disadvantaged groups. Features such as school_state, poverty_level, and teacher_teach_for_america could be proxies for socioeconomic or geographic disparities and thus require careful scrutiny. The audit should assess whether the model’s predicted probabilities systematically disadvantage certain groups, particularly among our focus of projects that were not fully funded. This involves analyzing the distribution of predicted probabilities across subgroups for the bottom 10% of predicted unfunded projects. A high representation of certain groups could indicate a systematic bias - which would help larger policy goals, as well as give insight as to how DonorsChoose should approach course correction in those subgroups.

The resolution of any bias we find can happen by employing post-processing techniques, or rework on the initial balancing of the data. This section in any machine learning project is important to substantiate by how much a data instance is away from ideal, as well as ensure that model training doesn't include inherent bias - which could inhibit its prediction capacity and generalization.

Evaluation:

In terms of recall as one of the metrics of focus, we realized we should correctly predict those outcomes where the project is funded. Of these, how many did we actually predict? Big numbers are good here. Smaller the false negatives, greater is the sensitivity, better is the model.

Precision is also important in our case. We must pick (predict) exactly the fully_funded = 1 projects. The greater the correctness of the predictions (predicting funded/ not funded exactly as projects are funded/ not-funded respectively), the higher this number. We would like to thus decrease the predictions as fully_funded = !T, especially when they are full_funded = t as ground truth. This will ensure that the content analyst focuses only on those projects which really need the attention.

The reasoning behind using the f-1 score as our metric of evaluation is based in the idea that it is representative of recall, precision and thereby able to evaluate how our models perform on classifying the projects as accurately as possible and in essence reducing false positives. False positives here mean the projects that *aren't* actually funded, but as classified as fully_funded = 1. This is problematic because these projects miss out on our eventual ranking of the worst performing projects based on their probability of being fully_funded = 1.

Important features:

We identified the most important features using the feature importance code already a part of the XGBoost pipeline. This list has been elaborated below. This was done to identify how we are/ aren't selecting consequential features and if there is any observable room for improvement. The idea was that if we perform reliably on the observables, it is more likely that we will perform reliably on unobserved data and unobservable covariates.

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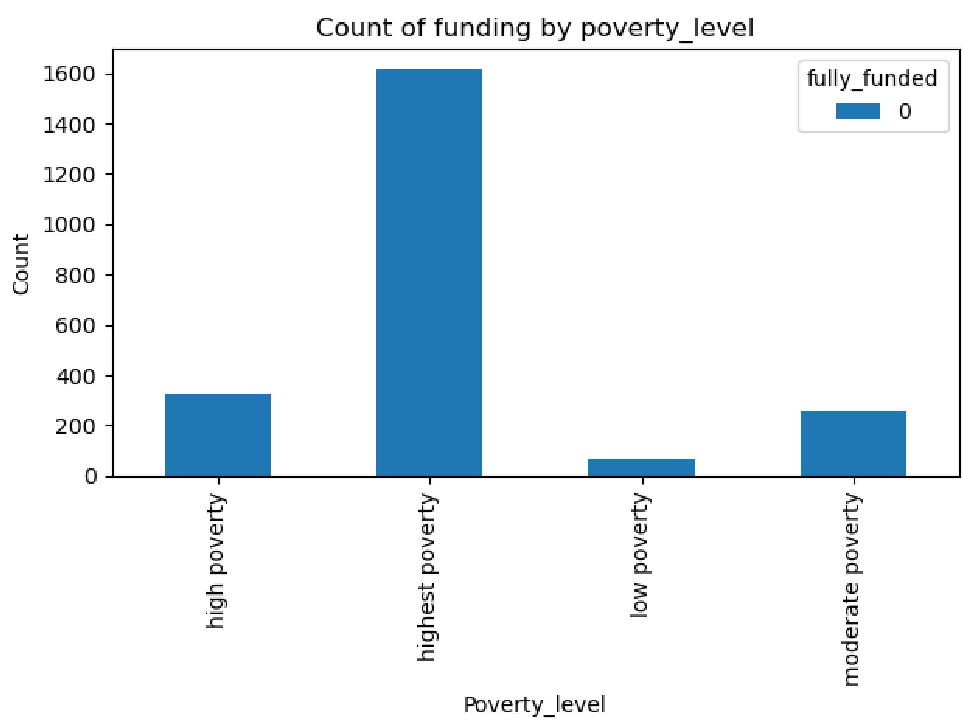
ohe__donor_state_Not available: 0.056183185428380966
ohe__school_charter_t: 0.05048522725701332
ohe__school_state_NY: 0.03988209739327431
ohe__resource_type_Technology: 0.038471899926662445
ohe__poverty_level_highest poverty: 0.03309936448931694
ohe__great_chat_t: 0.02811414748430252
ohe__vendorid_767.0: 0.024554502218961716
ohe__primary_focus_subject_Literature & Writing: 0.022262055426836014
ohe__primary_focus_subject_Literacy: 0.021610524505376816
ohe__primary_focus_subject_Special Needs: 0.020962076261639595
```

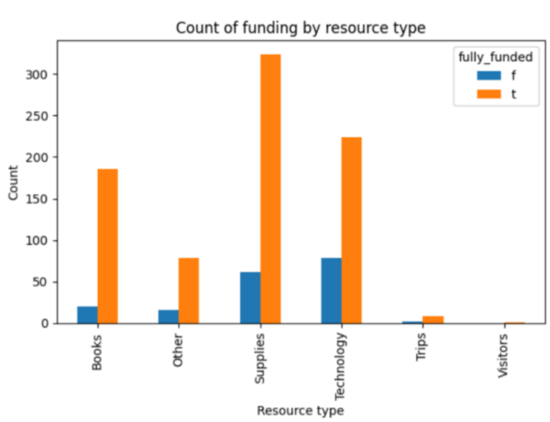
Analysis and Evaluations:

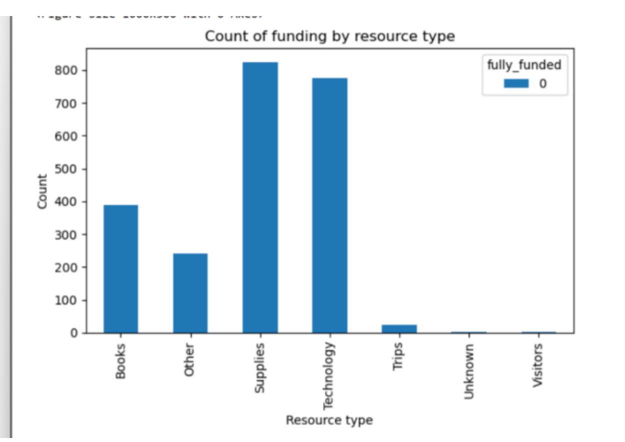
We find from our models that fully_funded = 1 is predicted with an accuracy of 0.81. In terms of the evaluation with the metric we have chosen - f-1 score for fully_funded = 1 is 0.89 and for fully_funded = 0, it is 0.32. This is considerably lower than our preference would suggest. We still go ahead and rank order projects with the lowest probabilities of being funded from the bottom up. The resource capacity is for 10% of the projects and this is what we constrain our analysis to. After identifying the worst performing projects, we find our critical set and run analysis over it. In terms of helping a supposed content analyst, we analyse how different aspects of the data relates with fully_funded = 0. This is to help identify the exact scopes where the analyst can work on per project to help them move to fully_funded = 1. This analysis is also generally reproducible to create a best practices readme for project posters to access before posting their projects.

In general, our analysis predicts fully_funded = 0 with an improvable f-1 score. We use these predictions to identify the bottom-most decile in terms of possibility of fully_funded = 0. The analysis is run on this, because it solves our initial problem of helping with the exact projects, in terms of the scope of a content analyst expert.

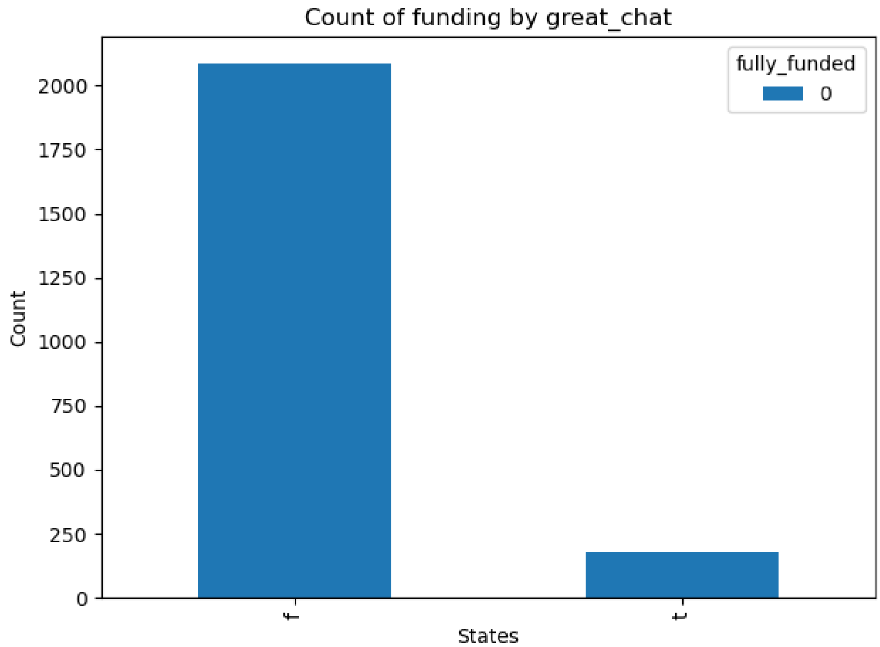


Projects are not funded parallely to the count of projects posted across categories, except moderate_poverty





Projects are not funded parallely to the varied counts of the respective categories, except 'Books.' 'Books' as a resource are requirements of lots of projects, but see disproportionate under-funding relative to other categories. Likewise, a disproportionately bigger number of predicted under-funded projects have the resource ask as 'Supplies.'



The devil is in the details.

One actionable insight would be ensuring that the content analyst works with predicted under-funded projects to improve project communications - so that the large representation of great_chat = f in under-funded projects is dropped. This also makes the case for a deeper look at unstructured data for every project like its descriptions and requirement statements. Maybe improving those can generally improve the projects funding by increasing donor interest.

Discussion on Evaluations

This project aims to reduce the funding disparities in U.S. public schools by maximizing the number of school projects funded through the DonorsChoose platform. We achieved this goal by predicting the bottom 10% of projects least likely to be fully funded, allowing DonorsChoose to intervene earlier and help the projects secure full funding. Using historical data from 2003 to 2010 to predict outcomes in 2011- and extending the model with 2012 data to predict 2013, we developed a machine learning pipeline that highlights the critical factors contributing to funding failure.

Our model finds that poverty level, donors' engagement with the campaign on the platform, and teacher's affiliation consistently influenced the likelihood of funding success. The gradient boosting algorithm predicted 1,934 projects as least likely to be fully funded in 2011. Notably, over 80% of those projects came from schools with the highest poverty level- meaning that 65% of their students qualified for free or reduced-price lunches. As highlighted in the introduction, schools in high poverty areas tend to be most impacted by funding disparities. The overrepresentation of these schools among the least likely to be funded emphasizes that, even on platforms designed to increase access to education funding, structural inequities persist. It's also important to note that the large proportion of such schools in the bottom 10% may partly reflect their larger presence among all funding requests, compounding the effect of both funding demand volume and systemic disadvantage.

Donor engagement, or the percentage of donors leaving unique comments, was also strongly associated with funding success. Our model shows that approximately 80% of projects that had below average donor engagement were among those least likely to be fully funded. Additionally, teacher affiliation emerged as a critical factor. Over 85% of projects submitted by teachers not affiliated with *Teach for America* were predicted to be unsuccessful. These findings point to the importance of personal and institutional engagement in driving donor behavior. Consistent with Alazazi et al.'s findings, our analysis suggests that positive engagement-whether through individual donor interactions or institutional affiliation could make a difference in which projects are successfully funded.

In 2013, the projects predicted to not be fully funded increased by 17%. The same predictive patterns hold, though we observe a 200% increase in the number of unsuccessful projects requesting technology resources. This trend likely reflects the higher cost associated with technology compared to traditional resources like books or supplies, making it more difficult for projects to meet their target funding.

While some factors such as the type of resource requested and geographic location of projects correlated with lower funding likelihood, their impact was less pronounced than the primary factors outlined above. These findings suggest that while regional or resource-based disparities do play a role, they are secondary to more consequential predictors such as poverty level, donor engagement, and teacher affiliation. These insights point to several policy changes that can be enacted both at DonorsChoose and teachers' levels to maximize funding.

Policy Recommendations

Increase Visibility of Projects Unlikely to be Fully Funded on DonorsChoose Platforms

DonorsChoose should highlight the projects predicted to not receive full funding on their social media platforms. For instance, every week, digital content experts should pick 10 projects in the bottom 10% and from schools in highly poor neighborhoods. They should then post about these projects on their Instagram, Facebook, and Twitter pages to increase donor engagement. Once they make their social media posts, they should encourage campaign leaders to leave positive messages and to share the post with their network. As we discuss above, projects that were predicted in the bottom 10% often lacked substantial donor interactions. The social media posts and comments can increase donor engagement and get the projects to their target funding. Namchul Shin's study on social media and charitable giving finds that the number of followers on Twitter is strongly associated with increased donations.[15] DonorsChoose can choose to focus their engagement efforts on Twitter to maximize donations.

Consolidate Projects Submission at the School Level to Increase Donations

DonorsChoose should implement a feature that allows teachers to submit multiple projects under a single school-level campaign. Through this system, DonorsChoose will not display individual, disconnected projects pages. Rather, campaigns from the same school will be highlighted under one profile. Donors will see the total funding goal for the school, with transparent breakdowns about where funds will be allocated. This strategy can maximize

[15] Shin, Namchul. "The Impact of Social Media on Charitable Giving for Nonprofit Organization." *Journal of International Technology and Information Management* 32, no. 1 (January 1, 2024): 122–40. https://doi.org/10.58729/1941-6679.1580.

donations especially given DonorsChoose's low retention rate of donors.[16] Behavioral Economics studies find that too much choice may increase confusion and decision deferral- a phenomenon known as the paradox of choice.[17]For example, studies found that when students were presented with more volunteering opportunities, they were more likely to defer making a decision.[18] Similarly, an overabundance of projects on DonorsChoose could discourage donors from selecting more than one or donate at all.

While some research have questioned the robustness of the choice overload effect, a flexible approach can address this uncertainty. One compromise would be to retain the current project-by-project structure but add an option for donors to "Donate to the school instead" when viewing an individual project. This strategy would simplify decision-making for donors and increase donations. Additionally, this structure could help mitigate the institutional affiliation bias found in our model, where teachers affiliated with organizations such as *Teach for America* were more likely to receive full funding. Consolidating the projects would make funding less about individual teacher networks and more about collective school needs.

Ultimately, these changes would reduce cognitive loads for donors, promote equity in funding across classrooms, and help close the gap between well-connected and under-resources schools.

Restructure U.S. Public School Funding to Close the Funding Gap

Platform-level interventions can reduce disparities in the short term but they cannot be a substitute for systemic change. Individual donations will never be sufficient to close the gap between school in more affluent neighborhoods and those in more impoverished communities. The U.S. federal and state governments should reevaluate public school funding formulas to ensure that resources are allocated based on student and community needs- not property tax revenue.

Limitations of Data

The data is highly imbalanced

- There isn't an equal distribution of the fully_funded variable, and 79% of the projects are funded. This makes training models difficult, because the majority class (here, fully_funded = 1) will be over-represented. Thus, we use SMOTE - in order to synthetically over-estimate the minority class (fully_funded = 0, here). This is a standard practise in machine learning methods that are used to classify new data.

No recency in data

- This data set is old, and a lot has changed in the non-profit space - with aspects of their functioning, like transparency measures, search optimization and operational characteristics like UI/UX, as well as project posting literacy, generally changing

[16] Dirisam, Joel Varma, Doina Bein, and Abhishek Verma. "Predictive Analytics of Donors in Crowdfunding Platforms: A Case Study on Donorschoose.Org." In 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC), 0834–38, 2021. https://doi.org/10.1109/CCWC51732.2021.9376094.

[17] Carroll, Lauren S., Mathew P. White, and Sabine Pahl. "The Impact of Excess Choice on Deferment of Decisions to Volunteer." *Judgment and Decision Making* 6, no. 7 (October 2011): 629–37. https://doi.org/10.1017/S1930297500002667.

[18] Ibid.

significantly. For that recent, the ML methods built in the project are not deployable in the real world - in 2025. Greater recency in data will help account for such changes in the non-profit ecosystem and this will help prediction via classification more significantly.

- A data aspect that wasn't present, but would help efficiency versus equity questions related to the problem goals is information about funding levels. This will help identify fully_funded = 0 projects that specifically fall just short of the fully_funding = 1 threshold. This identification can help such projects cross the threshold with a little bit of help, spread over a lot of projects. Currently, the content analyst goes over a few projects and has to work a lot to pull their performance. respective amounts(funding thresholds) needed by the various projects posted which could provide more insight into why some projects were funded while others weren't.

Limitations of the analysis

Limitations of ML methods

- Our project pipeline is built out in three stages as explained earlier. ML models are trained using three different ML methods on a dataset with 15,000 projects. This number was chosen to optimize for the computer available. Systematic sampling was used in any case, but greater EDA in the earlier stages could possibly help identify a more scientific approach to this issue.
- Another issue with the analysis is based on the pipeline stages. Since sklearn's predictor predicts probabilities greater than the chosen threshold, our trained ML methods are essentially predicting projects where fully_funded = 1. The evaluation metric is based on =1 predictions. Our final prediction implementation using the best performing ML method as a saved model architecture is for fully_funded = 0. This makes the metric evaluation comparison less reliable than ideal.
- Our analysis doesn't include unstructured data components directly but categorical variables derived from them('great_chat' feature was derived from the 'donation_message' feature). That is to say, for the lack of adequate computing resources, we overlooked (dropped) data about project descriptions - which in an ideal scenario would really help the predictions we make for a content analyst. We have factored in categorical variables that simulate the essence of these descriptive details for every project, like 'great_chat' but not to the extent of the information we miss out on due to using objective variables only.
- A lot of the analysis is based on total counts - which have as such increased every year. So it would be natural to see increased counts of projects not being funded in NY and CA - as the number of projects in the states are high as such. Percentages would be a better measure in such cases.

Scope of improvement

Data-related

- Data updates, while not possible for this case, could allow for additional insight on what is happening for projects that are funded, as well as are not funded and what is the distance between their average characteristics. This we already see with the 2003-2011 model predictions versus the 2003-2013 model prediction. Greater data depth in the years

2012-13 have allowed predictions to better by margins of 2% in accuracy. Should we be able to access the depth of data generated since 2013, the predictive powers would increase for the models.

Model architecture and analysis

- Using Pittsburgh supercomputer and Google Collab's free compute on student accounts to access computing power that would enable the training to happen on the entire data-set instead of systematic sampling only.
- Re-work the architecture of the model so that fully_funded = 0. This would overcome the predictor limitation on identifying p > 0.5/ threshold as it would now work *for* the prediction of fully_funded = 1 - in this case that meaning projects that are not funded. This simple trick could overcome the metrics comparison equivalency issue.
- Include sentiment analysis to score descriptive variables as positive, negative or neutral and use such variables as predictors of y.
- Support-vector-machines, a supervised learning method used for classification, has been used by a lot of similar projects that we researched about. Using this method as one of the ML method options would improve the choice for analysts - who can decide from a larger set of options.

Proposal for Future Work

While our study focuses on identifying the bottom 10% of projects least likely to be funded , a future study could shift attention to the top 10% of underfunded campaigns- those that are just below the funding threshold. These near-miss projects may require minimal additional support to become fully funded, making more cost-effective targets for DonorsChoose intervention. This strategy could help DonorsChoose allocate resources more efficiently, as smaller efforts such as targeted promotion or donor matching might be sufficient to get these projects to the finish line. However, unlike our current equity-driven approach, this method may not prioritize schools in high poverty areas. Consequently, it would involve a different set of equity and efficiency tradeoffs that should be carefully reviewed.

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